

DSAA 1001 Week 4 Validation Set

Limit Theorems, Information, and Streaming Algorithms

Instructions

- Purpose: Self-assessment and practice before test
- Time: 30 minutes | Total: 120 pts
- Open book, AI tools allowed
- References: Learning Sheet 1, Sections 9-10; Learning Sheet 2, Sections 1-6

Part A: Multiple Choice [40 points]

Question 1. (5 pts) Independent random variables X and Y have variances 4 and 9. What is $\text{Var}(2X - Y)$?

- a) 5
- b) 13
- c) 20
- d) 25

Answer: _____

Question 2. (5 pts) Among distributions on the same finite support, which statement about entropy is correct?

- a) The uniform distribution has maximum entropy
- b) A certain outcome has maximum entropy
- c) Entropy is a directed distance between two models
- d) Entropy always increases when one outcome becomes more likely

Answer: _____

Question 3. (5 pts) What makes a streaming summary mergeable?

- a) It stores every raw item from every monitor
- b) It answers only after the stream ends
- c) Summaries can be combined without replaying the raw streams
- d) Its queries are necessarily exact

Answer: _____

Question 4. (5 pts) If r independent unbiased estimators each have variance σ^2 , what is the variance of their average?

- a) $r\sigma^2$
- b) $\frac{\sigma^2}{r}$
- c) σ^2
- d) $\frac{\sigma}{r}$

Answer: _____

Question 5. (5 pts) At time $t + 1$ in reservoir sampling, what is the probability that a particular occupied reservoir slot is replaced?

- a) $\frac{k}{t+1}$
- b) $\frac{1}{k}$
- c) $\frac{k}{t}$

d) $\frac{1}{t+1}$

Answer: _____

Question 6. (5 pts) What is the main trade-off when the base of a Morris counter is decreased from 2 toward 1?

- a) The estimate has finer resolution, but the stored counter grows faster
- b) The estimate becomes exact with no memory cost
- c) The counter uses less memory but gains more variance
- d) The update no longer uses randomness

Answer: _____

Question 7. (5 pts) Two distinct keys are mapped by a pairwise-independent uniform hash into m buckets. What is their collision probability?

- a) 0
- b) $\frac{1}{m^2}$
- c) $\frac{1}{m}$
- d) $\frac{2}{m}$

Answer: _____

Question 8. (5 pts) Why does increasing the number of Bloom-filter hashes not reduce the false-positive rate indefinitely when the bit-array size and number of inserted items are fixed?

- a) More hashes introduce false negatives
- b) Too many hashes fill much of the bit array, making all-one query patterns more common
- c) More hashes make the bit array smaller
- d) The false-positive rate never depends on the number of hashes

Answer: _____

Part B: Medium Question 9 [20 points]

Question 9. (20 pts) Let X and Y be independent Bernoulli variables with $P(X = 1) = P(Y = 1) = 0.4$, and let $Z = X + Y$.

- (a) [6 marks] Use convolution, or enumerate all contributing pairs, to derive the PMF of Z .
- (b) [4 marks] Let $B = \sum_{i=1}^{100} X_i$ for independent copies with the same distribution. Compute $E[B]$ and $\text{Var}(B)$.
- (c) [6 marks] Write both the exact expression for $P(B \geq 50)$ and its continuity-corrected CLT approximation in terms of Φ .
- (d) [4 marks] Explain what the LLN says about $\overline{X}_n$ and what the CLT adds. State one circumstance in which the Gaussian approximation can be poor.

Part C: Medium Question 10 [20 points]

Question 10. (20 pts) A base-2 Morris counter returns $\hat{t} = 2^{C_t} - 1$. A counter is observed at $C_t = 7$, and the true stream length is $t = 100$. You may use $\text{Var}(\hat{t}) = \frac{t^2 - t}{2}$.

- (a) [4 marks] Compute the observed estimate.
- (b) [5 marks] Compute the theoretical variance and standard deviation of one estimate.
- (c) [6 marks] Nine independent counters are averaged. Compute the variance and standard deviation of the average, then use Chebyshev's inequality to bound the probability that the average misses 100 by at least 50.
- (d) [5 marks] Explain why unbiasedness alone does not make the observed estimate reliable and identify where independence is required in part (c).

Part D: Hard Question 11 [40 points]

Question 11. (40 pts) A network monitor records one indicator for each of 100 periods: $X_i = 1$ if period i contains a failure, and $X_i = 0$ otherwise. Let $S = \sum_{i=1}^{100} X_i$.

At the start of the day, the network is in one of two unobserved conditions. With probability 0.75 it is a **quiet day**; conditional on this condition, the X_i are independent Bernoulli variables with failure probability 0.04. With probability 0.25 it is a **busy day**; conditional on this condition, the X_i are independent Bernoulli variables with failure probability 0.20.

- (a) [8 marks] If the day condition is unavailable and the 100 indicators must be modeled as i.i.d. Bernoulli variables, find the failure probability for that model and state the resulting distribution of S . Call this model Q .
- (b) [8 marks] Under the true two-condition process, derive $E[S]$ by conditioning on the day type.
- (c) [16 marks] Derive the exact PMF of S under the true process as a mixture of two binomial PMFs. Write exact expressions for $P(S \geq 12)$ under the true process and under Q . For $B \sim \text{Binomial}(n, p)$, use the continuity-corrected approximation $P(B \geq k) \approx 1 - \Phi\left(\frac{k - 0.5 - np}{\sqrt{np(1-p)}}\right)$ to estimate both tail probabilities.
- (d) [8 marks] Consider one particular ordered sequence of 100 indicators containing exactly 20 failures. Write its probability under the true process and under Q , and form their likelihood ratio. Which

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model gives the larger likelihood? You may either compute the ratio or give a convincing argument without the exact computation.

End of Validation Set — Check answers with AI and discuss